REVIEW OF THE OFFICIAL REVIEWER

on the dissertation work of Yesmukhamedov Nurmaganbet Seitkaliuly on the topic «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy», provided for the degree of Doctor of Philosophy (PhD) in the educational program: 8D06102 — Computer and Software Engineering

№	**Criteria**	**Eligibility (please select one answer option)**	**Justification of the official reviewer's position**
1	The topic of the thesis (as of the date of its approval) corresponds to the directions of scientific development and/or government programs	1.1 Compliance with priority areas of scientific development or government programs: 1) The dissertation was completed within the project or target program financed from the state budget (indicate the name and number of the project or program); 2) The dissertation was completed within the other state program (indicate the name of the program); <u>3) The thesis corresponds to the priority direction of scientific development, approved by the Higher Scientific and Technical Commission under the Government of the Republic</u>	The topic corresponds to the priority direction of scientific development «Information and communication technologies» approved by the Higher Scientific and Technical Commission under the Government of the Republic of Kazakhstan, and to the state priorities of healthcare digitalisation. The documents to which the direction of the work corresponds are: 1. The Concept for the Development of Artificial Intelligence for 2024–2029. 2. The Address of the President of the Republic of Kazakhstan «Kazakhstan in the Era of Artificial Intelligence» of 8 September 2025. 3. The Law of the Republic of Kazakhstan «On Artificial Intelligence» No. 230-VIII of 17 November 2025. 4. Subparagraph 2 of paragraph 3 of Article 20 of the Law of the Republic of Kazakhstan «On Science», in accordance with which the work was carried out. The volume states, and I confirm, that the research was not financed from the state budget and was not performed under a state programme; what is claimed is a correspondence of direction, not a mandate. Considered from the side of application, the correspondence is close: diabetic retinopathy is among the leading

		of Kazakhstan (indicate direction).	causes of preventable blindness in the working-age population, screening coverage in rural and underserved regions is limited by the number of ophthalmologists rather than by the availability of cameras, and the national eHealth platform is the natural point of integration for a screening service of this kind.
2	Importance for science	The work makes / does not make a significant contribution to science and its importance is well-disclosed / not disclosed.	The contribution is significant and, in my reading, its centre of gravity lies in what the work establishes about behaviour outside the training domain. The literature routinely reports in-domain gains from image enhancement and routinely leaves the external consequences of those gains to assumption. Here the question is put directly and separately on seven corpora beyond the training set, acquired with cameras of four manufacturers, and the answer is reported corpus by corpus without aggregation into a single figure of merit. Alongside this, the mechanism usually invoked to explain such behaviour — the reduction of the distance between the training distribution and the target distribution — is measured rather than asserted, at the penultimate layer, under the mandatory condition that normalisation uses source-domain statistics and is never recomputed on the target, so that the measured quantity is a property of the preprocessing and not of a procedure fitted to the target. The importance of the work is disclosed accurately: the volume states what the external results do establish and, with equal clarity, what they do not.
3	The principle of self-reliance	Self-reliance level: 1) high; 2) average; 3) low; 4) no independence.	The candidate designed, executed and analysed the whole experimental programme, including the external evaluations on corpora he did not create, and the volume distinguishes his own contribution from prior work where the two touch. Two things persuade me of genuine independence rather than

			competent execution. First, the acceptance gate applied to the initialisation of the integrated arm: rather than assuming a pretraining strategy, the candidate tested candidate initialisations with a frozen-backbone linear probe and reports that label-free self-supervision trained from scratch on the in-domain corpus failed that gate and was not admitted. A researcher working to a predetermined conclusion does not publish that. Second, the analytic identification of a defect shared by three robustness measures widely used in this literature — a finding that removes a convenient way of presenting his own external results and is stated as strictly descriptive, rehabilitating none of them. Of the five publications on the topic, the candidate is first author of two.
4	The principle of internal unity	4.1 Justification of the relevance of the thesis: 1) justified; 2) partially justified; 3) not justified.	The relevance is justified from both sides — the clinical need for screening coverage under a shortage of specialists, and the methodological gap in the treatment of preprocessing — and neither side is left as a general appeal. The clinical case is made in terms a service operator would recognise: asymptomatic early stages, a growing diabetic population, and a bottleneck that is human rather than instrumental. The methodological case names a specific absence and then addresses it.
		4.2 The content of the thesis reflects its topic: 1) reflects; 2) partially reflects; 3) does not reflect.	The content answers the topic as stated, and it answers it across the range the topic implies rather than on the training corpus alone: enhancement and classification are treated as one object, and the object is then followed onto seven further corpora. A volume that stopped at the in-domain contrast would still match the title; this one goes considerably further.
		4.3 The purpose and objectives correspond to the topic of the thesis:	The aim and the six objectives correspond to the topic and to each other: analysis of the problem domain, theoretical foundations, formalisation of the methodology, the experimental programme, the

		1) correspond; 2) partially correspond; 3) do not correspond.	reliability apparatus, and the screening architecture. The last objective is the one most easily left decorative in a technical dissertation, and it is not: the architecture is specified against the functional and non-functional requirements the earlier chapters establish.
		4.4 All sections and provisions of the dissertation are logically interconnected: 1) completely interconnected; 2) the relationship is partial; 3) there is no relationship.	The internal unity is carried by the dataset architecture as much as by the chapter sequence: the eight corpora are organised into functional tiers, each tier answering one question — primary training and ablation; cross-dataset transfer; explainability against pixel-level annotation; external clinical evaluation; device domain shift; training under data scarcity — and no corpus is asked to answer two questions without that being declared. Where two of the five camera groupings coincide with the external clinical corpora, the volume says so and declines to count them as independent replication.
		4.5 The new solutions (principles, methods) proposed by the author are argued and evaluated in comparison with known solutions: 1) there is a critical analysis; 2) partial analysis; 3) the analysis does not represent one's own opinions, but quotes from other authors.	The critical analysis in Chapters 1 and 5 is the author's own: existing systems are compared and their limitations argued, and the work is deliberately placed against them without being ranked among them, since no two of the compared results share task definition, corpus, reference standard and metric at once. The same discipline extends to the author's own measures: three robustness measures in common use are examined analytically and found to share a structural defect, which is a critical evaluation of the field's instruments rather than of its results.
5	The principle of scientific novelty	5.1 Are the scientific results and provisions new? 1) completely new;	Among the twelve elements of novelty declared in the volume I regard the following as the most substantial. The Attention–Lesion Overlap metric, introduced as a primary and deliberately

		2) partially new (25–75% are new); 3) not new (less than 25% are new).	asymmetric measure of explainability: it reports the fraction of an expert-annotated lesion that is covered by model attention, which is the quantity of interest for a screening reader, and it is complemented by Intersection-over-Union as a secondary and symmetric check; the volume is careful to describe what the metric establishes — alignment between model evidence and expert annotation — and not to call it localisation of pathology. The direct measurement of source-to-target distance at the penultimate layer across six external corpora, at the cost of forward passes only, with the falsifiability of the measurement recorded before it was made: two stages of the pipeline are bound to the source domain by construction, so a reversal was a live possibility rather than a rhetorical one. The validation of the pipeline across a tiered multi-corpus, multi-device architecture spanning four camera manufacturers, with corpora annotated under multi-disease taxonomies used as sources of diabetic retinopathy labels only. The analytic identification of the structural defect shared by the degradation quantity, the generalisation ratio and the retention ratio. And the eight-stage pipeline itself, formalised as a component of the model with the field-of-view mask carried into the network as a fourth input channel. Two contributions are non-empirical by design and the volume declares them as such: the coupled thermal-optical model of laser exposure is theoretical and computational, and the screening architecture is a design contribution.
		5.2 Are the conclusions of the dissertation new? <u>1) completely new</u>; 2) partially new (25–75% are	The conclusions that matter most for application are new. That the between-grouping dispersion of performance contracts under the integrated configuration — that the weakest camera grouping gains most — is a conclusion about the shape of the performance

		new); 3) not new (less than 25% are new).	distribution and not about its mean, and it is the shape that determines what a screening service can promise. That the reduction of source-to-target distance is present everywhere while its magnitude tracks no performance gain is a conclusion that constrains how the mechanism may be cited, and the volume draws it against its own interest.
		5.3 Technical, technological, economic, or managerial decisions are: 1) new and justified; 2) completely new; 3) partially new (25–75%); 4) not new (less than 25%).	The proposed screening architecture is a new technological and managerial solution: a configurable preprocessing engine, an inference module, store-and-forward telemedicine for intermittent connectivity, physician-in-the-loop decision support, and integration with PACS/EHR systems and the national eHealth platform. It is justified against stated functional and non-functional requirements and against a computational envelope a regional facility can meet, and it is submitted as a design specification rather than as an implemented system.
6	Validity of the main conclusions	All key findings are based / are not based on scientifically sound evidence or are reasonably well founded (for qualitative research and areas of training in the arts and humanities).	The evidential base is adequate to the conclusions drawn from it. The primary experiments rest on 5-fold patient-level stratified cross-validation with leakage control over approximately 35,126 graded images, with results reported as mean ± standard deviation on four metrics; the external evaluations rest on corpora of 413 and 1,744 images respectively, with differences reported together with their bootstrap confidence intervals and their p-values rather than as point estimates. Hypothesis testing uses McNemar and DeLong, correction for multiple comparisons uses Holm/Bonferroni, and variation across folds is modelled with mixed effects. Criteria were fixed before the experiments that tested them, which is what permits the outcomes to be read as tests rather than as descriptions. Where a criterion is cleared by both arms — as with the generalisation threshold in the

			transfer experiment and the retention floor in the device experiment — the volume says so and locates the evidence in the comparison rather than in the threshold. That is the correct reading, and the fact that the author makes it himself strengthens my confidence in the remainder.
7	Main provisions submitted for defense	The following questions must be answered for each provision separately: 7.1 Has the position been proven? 1) proven; 2) rather proven; 3) rather not proven; 4) not proven. 7.2 Is it trivial? 1) yes; 2) no. 7.3 Is it new? 1) yes; 2) no. 7.4 Application level: 1) narrow; 2) medium; 3) wide. 7.5 Is it proven in the article? 1) yes; 2) no.	Eleven provisions are submitted; each is answered below in the order 7.1 / 7.2 / 7.3 / 7.4 / 7.5. **Provision 1.** Preprocessing as a component of the diagnostic model — the reframing from the end-to-end paradigm to the integrated paradigm. 7.1 rather proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. Methodological and non-empirical, and submitted at that strength.
			Provision 2. The integrated configuration dominates the baseline on EyePACS on both backbones under the conjunctive pre-registered criterion. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. The criterion is satisfied in all three of its components on both backbones, with no interaction with the architecture ($p = 0.31$). The provision concerns the configuration as a whole, the arms differing in initialisation as well as in preprocessing.
			Provision 3. Interior optima of the CLAHE parameters and of the flat-field σ within the ranges examined. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Bounded to the ranges swept and confirmed on held-out data; the values are properties of the corpus rather than portable constants.
			Provision 4. Separability, monotonicity and ordering of the stage contributions. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Established at the resolution of groupings only, with the two

			photometric stages leading and adjacent ranks lying within noise.
			Provision 5. Reduction of source-to-target distance on every external corpus. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. The distance fell on all six corpora with every interval excluding zero, the divergence of the channel histograms falling by roughly a third in the same direction. The provision is submitted in direction only, and correctly so: the size of a reduction does not track the size of any performance gain, and the claim is mechanistic, carrying no assertion about diagnostic quality or device compatibility.
			Provision 6. Transfer to APTOS 2019 without retraining, clearing $G \geq 0.85$. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. $G = 0.8976$ against 0.8577 for the baseline, weighted F1 0.6465 → 0.7354, $\Delta = +0.0889$ with the interval [+0.0681, +0.1197]. Both arms clear the threshold, so the evidence lies in the comparison.
			Provision 7. Closer alignment of attention with expert lesion annotation. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Established on all four annotated lesion types, on both measures, and robustly across the attention threshold. It is alignment, not clinical localisation, on one corpus and one fold, and the qualitative examination on the clinical corpus was not carried out.
			Provision 8. Performance maintained across camera groupings with the dispersion reduced. 7.1 proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. All five groupings clear the retention floor for both arms, and the substantive result is the contraction of the between-group dispersion of weighted F1 from 0.0306 to 0.0130 with an interval excluding

			zero, the largest gain falling on the weakest grouping. Two of the five groupings are the external clinical corpora themselves, three aggregate several camera models rather than a single device, and none of this constitutes device certification.
			Provision 9. Higher absolute external weighted F1 on both clinical corpora, each difference reaching the minimal clinically important difference with its interval excluding zero. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. 0.5938 → 0.6627 with Δ = +0.0689, interval [+0.0494, +0.0968], and 0.6282 → 0.6823 with Δ = +0.0541, interval [+0.0362, +0.0814]. Submitted as higher absolute external performance and expressly not as resistance to degradation: relative to their own in-domain levels the two configurations decline almost identically. See also comment 1 below on the margin on the second corpus.
			Provision 10. A structural defect shared by three robustness measures in common use. 7.1 proven · 7.2 no · 7.3 yes · 7.4 wide · 7.5 yes. Analytic, strictly descriptive, rehabilitating no result of this work.
			Provision 11. The coupled thermal-optical model of laser exposure and the modular screening architecture. 7.1 rather proven · 7.2 no · 7.3 yes · 7.4 medium · 7.5 yes. Theoretical and design contributions respectively, neither clinically validated nor prototyped nor field-tested. I note that the volume expressly withholds clinical validity, readiness for unsupervised deployment, device certification, localisation of pathology and superiority over any named published system.
8	The principle of reliability.	8.1 The choice of methodology justified or is it	The choice of methods is justified at each point where an alternative existed, and the description is detailed enough to be repeated: the

	Reliability of sources and information provided	described in sufficient detail? 1) yes; 2) no.	backbones and their training regime, the loss and its weighting, the input resolution, the fold construction, and an explicit ingestion protocol for images arriving from outside the training corpus — the last being the element a deployment would depend on and the one most often left unspecified.
		8.2 The results of the thesis work were obtained using modern methods of scientific research and techniques for processing and interpreting data using computer technologies: 1) yes; 2) no.	The methods are contemporary and appropriate: convolutional classification on two established backbone families with five-class grading, Focal loss with inverse-frequency weighting for the class imbalance, 512 × 512 input, patient-level stratified folds, gradient-based attention analysis against pixel-level annotation, and a distributional measure over penultimate-layer features. The empirical basis is documented corpus by corpus with acquisition device and function, and the corpora annotated under multi-disease taxonomies are used only as sources of diabetic retinopathy labels — a restriction that matters for the validity of the device-shift results and is observed.
		8.3 Theoretical conclusions, models, identified relationships and patterns are proven and confirmed by experimental research: 1) yes; 2) no.	Theoretical statements are confirmed experimentally rather than left as claims: the ablation reproduces the whole in-domain gain of +0.0655 under a single initialisation, the sweeps confirm their optima on held-out data, and image quality is evidenced independently of the classifier by contrast-to-noise ratio, entropy and SSIM. The mechanistic claim is confirmed in the same spirit — measured directly on six corpora, and then bounded to a statement of direction when the measurement failed to support more.
		8.4 Important statements are supported / partially supported / not supported by references to current and reliable scientific literature.	The referencing is current where the subject demands it and conservative in the clinical parts, and it is used to support rather than to decorate: where the volume declines to rank itself among published systems, it gives the methodological reason and the sources against which the comparison was attempted.

		8.5 The literature sources used <u>are sufficient</u> / are not sufficient for the literature review.	The reference list of 107 sources is sufficient and appropriate to the subject. The volume also states three limits of its own apparatus — the per-experiment scope of the multiple-comparison correction, the single-fold basis of several evaluations, and the non-redistributable clinical corpus which makes one result externally irreproducible — and it distinguishes calibration, which it measures and reports, from any warrant of clinical decision reliability. Stating these raises rather than lowers my assessment of the work's reliability.
9	Principle of practical value	9.1 The dissertation has theoretical significance: <u>1) yes</u>; 2) no.	The theoretical significance is twofold: the reframing of preprocessing as a model component with the formalisations it required, and the conversion of an explanatory assumption into a measurable and falsifiable quantity. The second is the more portable of the two — the protocol condition under which the distance is computed can be applied by others to preprocessing regimes that have nothing to do with this one.
		9.2 The thesis has practical significance and there is a high probability of applying the obtained results in practice: <u>1) yes</u>; 2) no.	The practical value is, in my view, the strongest side of this work, and it rests on two things that are rarely present together. The first is that the pipeline is fully specified — stage by stage, with operators, parameters and order of application, its implementation reproduced in an appendix — and runs within an envelope a regional facility can meet: a single 12 GB accelerator, batch size 16, 512 × 512 input. The second is that its behaviour has been characterised outside the training domain, on external clinical corpora and across camera groupings, which is precisely the evidence a service operator needs before considering deployment; and the contraction of the between-group dispersion of performance is operationally more valuable than the average gain, because it is the weakest camera grouping that determines the quality of a screening service.

		9.3 Are the suggestions for practice new? <u>1) completely new</u>; 2) partially new (25–75%); 3) not new (less than 25%).	The modular architecture — configurable preprocessing engine, inference module, store-and-forward telemedicine for intermittent connectivity, physician-in-the-loop decision support, PACS/EHR and national eHealth integration — is directly applicable to screening in rural and underserved regions of Kazakhstan and is new as a whole. It is nevertheless a design specification: no prototype was implemented, no deployment was tested clinically, the data-protection provisions are alignment by design and not certified compliance, and the system is conceived as decision support in which the clinician retains the diagnosis and the responsibility for it. The parameter values are properties of the corpora on which they were fixed and would require re-derivation for a new acquisition setting.
10	Quality of writing and design	Quality of academic writing: <u>1) high</u>; 2) average; 3) below average; 4) low.	The volume is set out on 265 pages, contains 42 tables, 26 figures and two diagrams, and draws on 107 sources; the notation is consistent, the abbreviations are declared in a dedicated register, the illustrations are legible and numbered in document order, and the formatting conforms to the applicable state standard. What deserves particular mention is the discipline of the language in which results are reported. Each closing statement carries its qualification inline; external results are reported per corpus and never merged into a single number; where a threshold is cleared by both arms the text says so instead of presenting the pass as a discriminating result; and the distinction between what was measured, what was designed and what was modelled theoretically is maintained consistently. A reader of this volume is not at risk of taking away a stronger claim than the evidence supports, which is not a common quality.
11	Comments and		*1. On the external clinical evaluation, the margin by which the*

	suggestions on the dissertation		*difference on the second corpus exceeds the minimal clinically important difference is 0.0041 (Δ = 0.0541 against a threshold of 0.050), and the lower bound of the corresponding interval, +0.0362, lies below that threshold. The criterion adopted requires Δ ≥ MCID together with a lower bound above zero and is therefore satisfied, and I do not dispute the verdict; but the pass on this corpus rests on the form of the criterion rather than on a comfortable margin. This is stated in the limitations, and it should be stated once more, in a single sentence, at the point in the chapter where the result is first reported.* *2. The correlation between the measured reduction of source-to-target distance and the observed transfer gain across the external corpora is weak (ρ ≈ 0.49), so the mechanistic result supports a claim of direction and not of magnitude. The volume states this correctly in the qualification attached to the corresponding provision, but the chapter that reports the measurement concludes without restating it. Since this is the chapter a reader will consult when citing the mechanism, the direction-only reading should be repeated in its own conclusion.* *3. The programme of the explainability experiment declares a qualitative examination of attention maps on the Kazakh clinical corpus, and that examination was not carried out; the quantitative half against pixel-level annotation was completed in full and supports the corresponding provision. The absence is disclosed, which I note with approval, but it is disclosed later than it is introduced. It would be preferable to mark it as future work at the point where the qualitative component is first announced, so that the reader is not left expecting a result that does not follow.* The comments listed above concern the presentation of results

			rather than their substance, do not affect the validity of the conclusions, and do not alter my overall positive assessment of the dissertation.
12	The scientific level of the doctoral candidate's articles on the research topic (in the case of defending the dissertation in the form of a series of articles, the official reviewers will comment on the scientific level of each article)		The results of the dissertation are published in five scientific works corresponding to its topic. 1. An article in a journal indexed by Scopus — *Eastern-European Journal of Enterprise Technologies*, 2025, Vol. 4, No. 9(136), pp. 79–88, quartile Q3, CiteScore 2025 = 2.5, 42nd percentile in the category Computer Science Applications (#590 of 1022) and 44th percentile in Control and Systems Engineering (#217 of 390). 2. A paper in Scopus-indexed international conference proceedings — *Procedia Computer Science*, 2025, Vol. 272, pp. 496–501, presented at the 3rd International Workshop on Digital Society (DS 2025, Istanbul, Türkiye). 3–5. Three articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education of the Republic of Kazakhstan: *News of the National Academy of Sciences of the Republic of Kazakhstan, Physico-Mathematical Series*, 2025, Vol. 2(354), pp. 74–91; *Herald of the Kazakh-British Technical University*, 2025, No. 4(75), Vol. 22, pp. 119–130; *Herald of KazUTB*, 2025, Vol. 2, No. 27-740. The publications cover the principal directions of the dissertation — fundus image enhancement, preprocessing and analysis methods, the architecture of a healthcare information system, and the mathematical modelling of laser exposure on fundus tissues — and the results reported in them agree with those presented in the volume. Their scientific level corresponds to the requirements established for the defence of a doctoral dissertation.
13	The decision of the		To petition the Committee for the award of the degree of Doctor of

	official reviewer (according to paragraph 28 of the current Model Regulation)		Philosophy (PhD).

Conclusion

The presented dissertation for the degree of Doctor of Philosophy (PhD) by **Yesmukhamedov Nurmaganbet Seitkaliuly** on the topic «Computer System for Processing and Analysis of Eye Data to Support Laser Coagulation in the Treatment of Diabetic Retinopathy» is a completed independent piece of scientific research, which meets the requirements of the Rules for awarding degrees, and its author deserves to be petitioned before the Committee for the award of the degree of Doctor of Philosophy (PhD) in the educational program 8D06102 — Computer and Software Engineering.

Official reviewer:

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